

- A function $u : \mathbb{R}^L \rightarrow \mathbb{R}$ is **quasi-concave** if $\forall x, y \in \mathbb{R}^L$ and $\forall \alpha \in [0, 1]$, $u(\alpha x + (1 - \alpha)y) \geq \min\{u(x), u(y)\}$.
 - It is **strictly quasi-concave** if $\forall x, y \in \mathbb{R}^L$ with $x \neq y$ and $\forall \alpha \in (0, 1)$, $u(\alpha x + (1 - \alpha)y) > \min\{u(x), u(y)\}$.
 - It is **concave** if $\forall x, y \in \mathbb{R}^L$ and $\forall \alpha \in [0, 1]$, $u(\alpha x + (1 - \alpha)y) \geq \alpha u(x) + (1 - \alpha)u(y)$.
 - It is **strictly concave** if $\forall x, y \in \mathbb{R}^L$ with $x \neq y$ and $\forall \alpha \in (0, 1)$, $u(\alpha x + (1 - \alpha)y) > \alpha u(x) + (1 - \alpha)u(y)$.
- 1) For a C^2 function $u(x)$:
 $u(x)$ is concave $\iff$ its Hessian matrix $[\partial^2 u(x)/\partial x_i \partial x_j]$ is negative semidefinite for all x .
The Hessian matrix negative definite for all $\implies$ then it is strictly concave.
- 2) (Strict) Concavity $\implies$ (Strict) quasiconcavity.
- 3) Quasiconcavity (NOT concavity) is preserved by monotonic transformation, i.e., if $f(\cdot)$ is (strictly) quasi-concave, g is strictly increasing, then $g \circ f$ is (strictly) quasiconcave. Proof: $\forall x, y$ in the domain with $x \neq y$ and $\forall \alpha \in (0, 1)$, $g(f(\alpha x + (1 - \alpha)y)) > g(\min\{f(x), f(y)\}) = \min\{g(f(x)), g(f(y))\}$.
- An $n \times n$ matrix A is **positive (negative) definite** if for each $0 \neq x \in \mathbb{R}^n$, $x^T A x > 0$ (< 0); it is **positive (negative) semi-definite** if $x^T A x \geq 0$ (≤ 0).
 - The **k -th naturally ordered principal minor** (or **k -th order leading principal minor**) of A , A_k is a submatrix of A obtained when only the first k rows and columns are retained.
 - (Sylvester's criterion). Let A be a symmetric $n \times n$ matrix;
 1. A is negative definite $\iff (-1)^k |A_k| \geq 0$ for each $k \in \{1, 2, \dots, n\}$;
 2. A is positive definite $\iff |A_k| \geq 0$ for each $k \in \{1, 2, \dots, n\}$.
where $|A_k|$ is the determinant of k -th naturally ordered principal minor of A .
 - Theorem. Let A be a symmetric $n \times n$ matrix.
Let $\mathcal{A}_k$ be the **k -th order principal minor** (a submatrix of A when $n - k$ rows and the same $n - k$ columns are deleted) of A .
 1. A is negative semidefinite $\iff (-1)^k |\mathcal{A}_k| \geq 0$ for each $k \in \{1, 2, \dots, n\}$;
 2. A is positive semidefinite $\iff |\mathcal{A}_k| \geq 0$ for each $k \in \{1, 2, \dots, n\}$;